Corporate Finance

Lecture 6 extras: Spurious correlations, and a trader at La Défense

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A. Spurious correlations: data snooping in action

Remember data snooping?

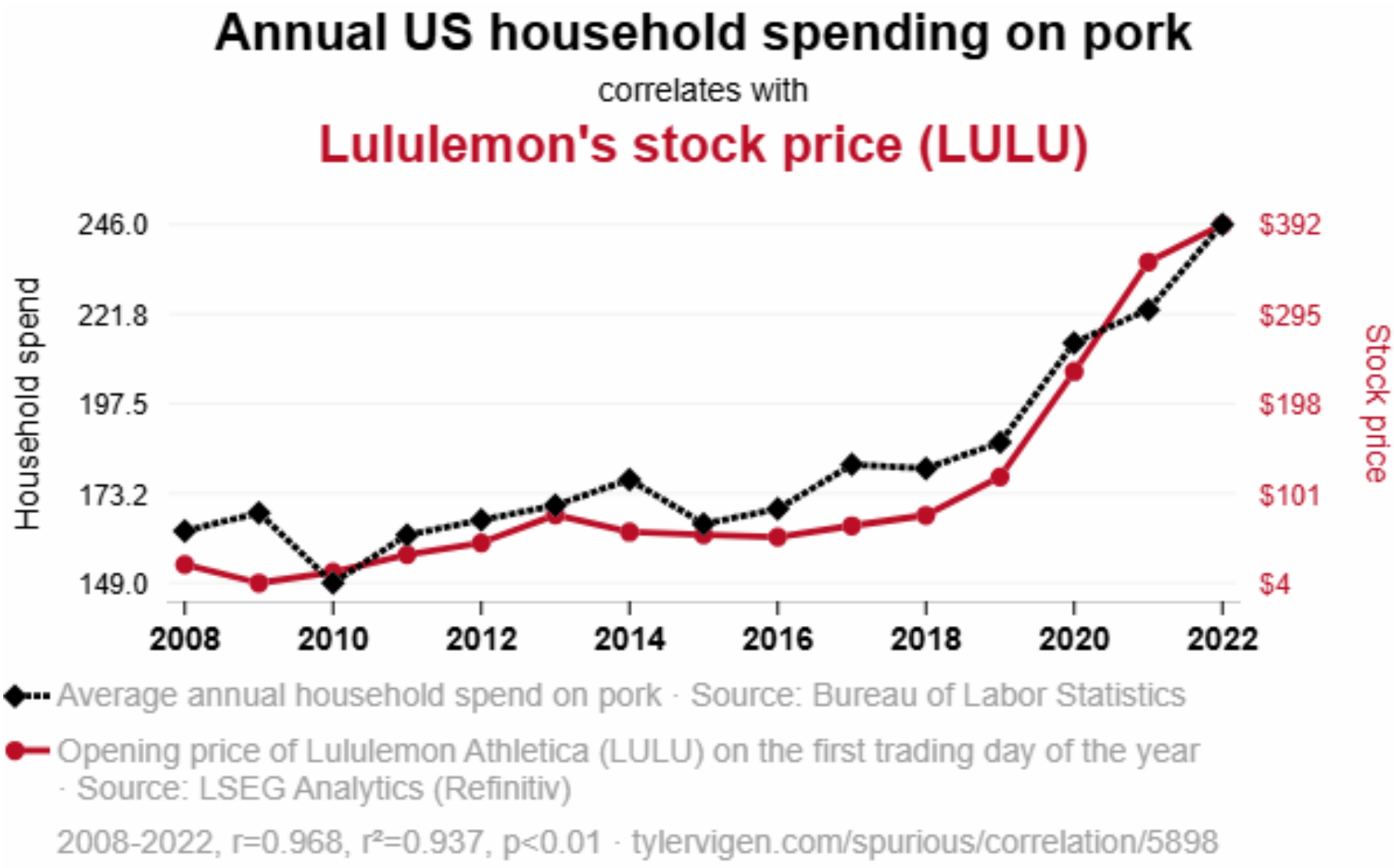
- If you test **hundreds** of characteristics on the same past data some will "beat the market" by pure luck. That is **data snooping**.
- When evaluating statistical relations on several data sets, some patterns will apper by luck.
- The website **Spurious Correlations** (tylervigen.com) does this on purpose: it keeps the pairs of series with a correlation close to 1. Reminder from Lecture 2:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \, \sigma_Y}$$

  The covariance rescaled by the two volatilities, so it lies between -1 and $+1$.

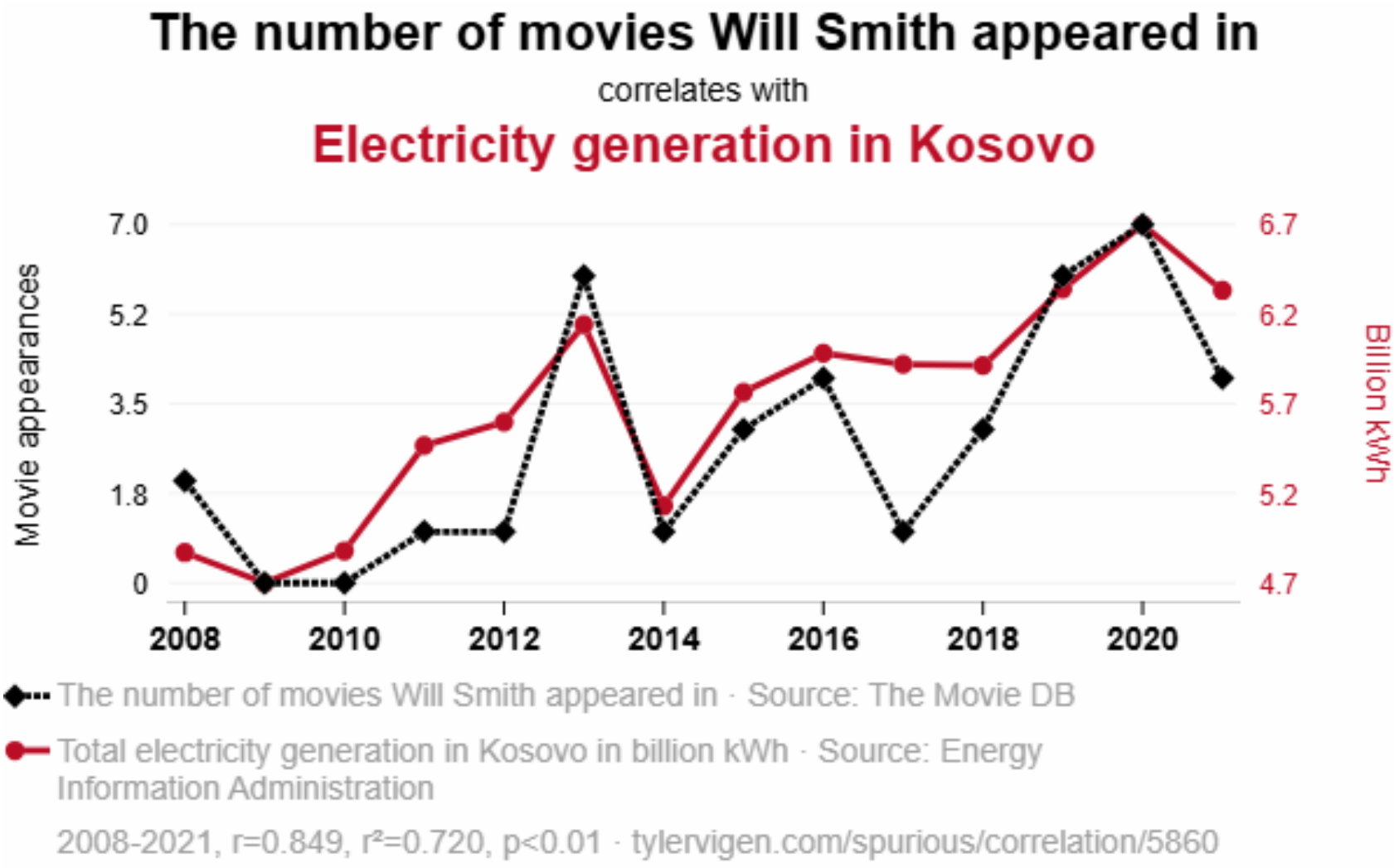
Search long enough and you will always find a pattern. Finding one proves nothing about the future.

Pork spending and Lululemon



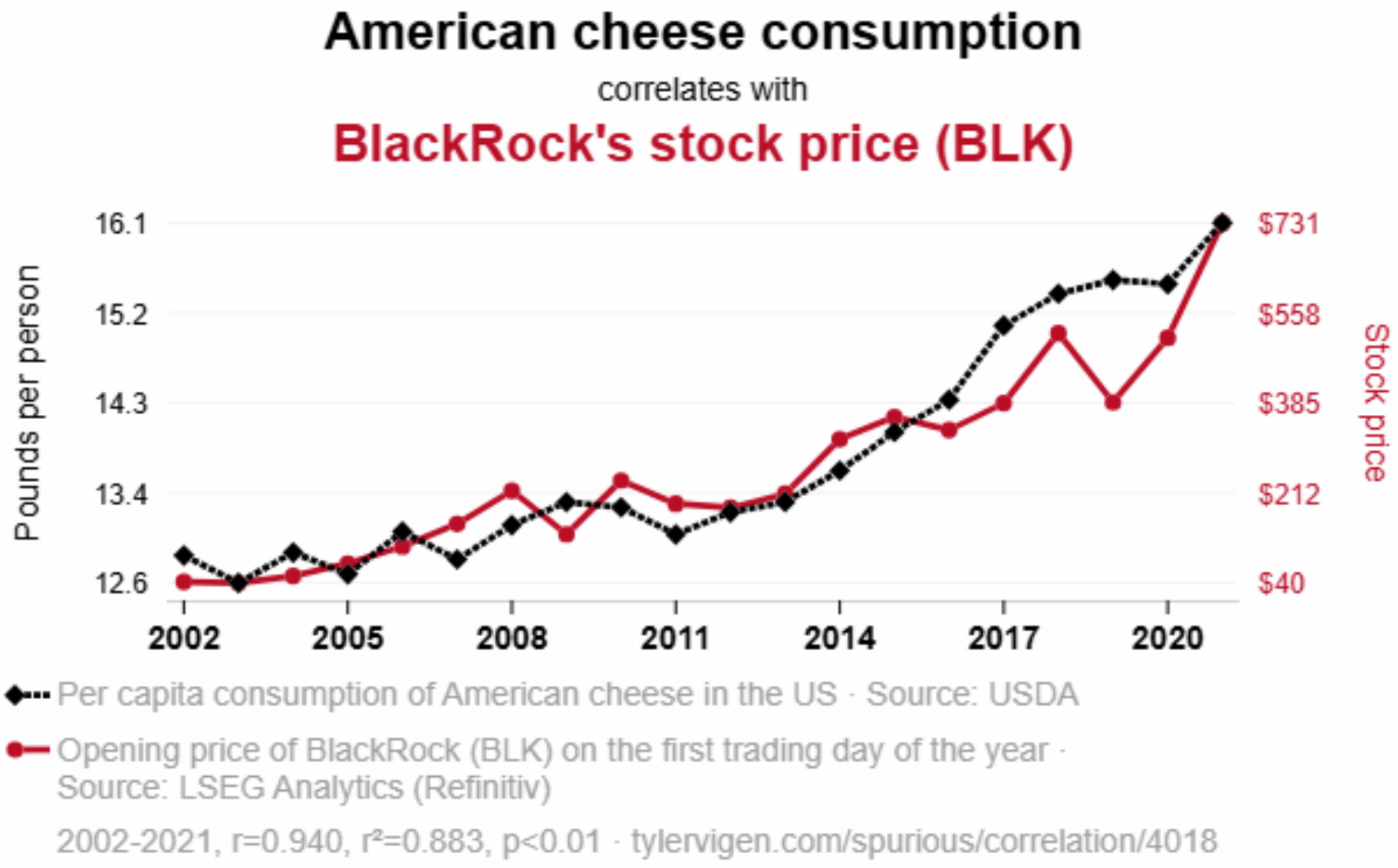
- Correlation **0.97**.

Will Smith and electricity in Kosovo



- Correlation of **0.85**.

American cheese and BlackRock



- Correlation of **0.94**. Both series simply grow over time, so they look alike.

B. L'Outsider: one trader, 50 billion euros

The Kerviel case in one slide

- **Jérôme Kerviel**, a junior trader at **Société Générale** in La Défense. His job: small, low-risk bets on European stock indices.
- In 2007 he secretly built up bets worth about **EUR 50 billion**, hidden with fake trades that cancelled them out on paper.
- January 2008: the bank discovers the positions and sells them in a few days, in a falling market. The loss: **EUR 4.9 billion**.
- 2010: he is sentenced to prison and ordered to repay the EUR 4.9 billion. 2016: an appeal court cuts that to EUR 1 million, because the bank's own controls had failed.

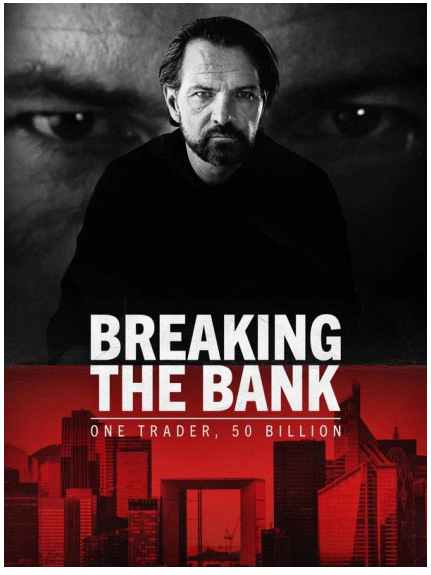
50 billion is what was **at risk**. 4.9 billion is what was actually **lost**.

How big is EUR 50 billion?

- **More than the whole bank.** In 2007, all of Société Générale's shares together were worth **EUR 43.16 billion** (its market capitalization), less than the bets of this one trader.
- **More than the yearly output of whole countries.** EUR 50 billion (about USD 54 billion) is more than everything produced in one year (GDP, 2024) in:
  - **Estonia**: about USD 43 billion
  - **Cyprus**: about USD 37 billion
  - **Iceland**: about USD 34 billion

The film and the documentary





- ***L'Outsider*** (2016), by Christophe Barratier, with Arthur Dupont as Kerviel. Tagline: "You can cheat to win, not to lose."
- ***Kerviel: un trader, 50 milliards*** (*Breaking the Bank: One Trader, 50 Billion*, 2024): a four-part documentary on HBO Max, with Kerviel and his former bosses.